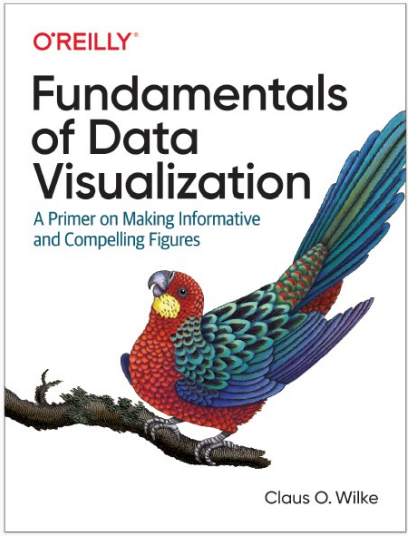
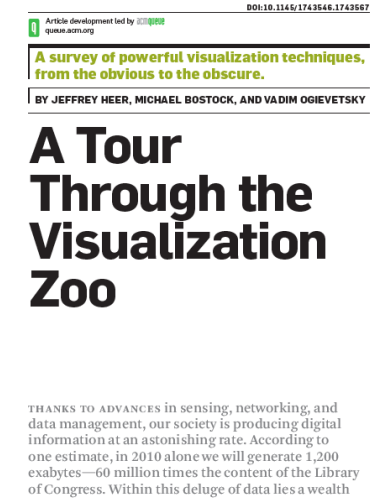
Introduction to Data Visualization

Veera Muangsin

References

- Book: Fundamentals of Data Visualization, Claus O. Wilke, 2019. clauswilke.com/dataviz
- Book: Visualization Analysis & Design, Tamara Munzner, CRC Press, 2014.
- Book: Storytelling with Data, Cole Nussbaumer Knaflic, Wiley, 2015.
- Book: How Charts Lie, Alberto Cairo, W. W. Norton, 2019.
- Article: A Tour Through the Visualization Zoo, Heer, Bostock & Ogievetsky, CACM, 2010. doi:10.1145/1743546.1743567





Examples

https://public.tableau.com/app/discover/viz-of-the-day
https://community.fabric.microsoft.com/t5/Data-Stories-Gallery/bd-p/DataStoriesGallery
https://www.gapminder.org/tools/
https://flowingdata.com/
https://truth-and-beauty.net/
https://www.dataviz-inspiration.com/
https://www.data-to-viz.com/

What is Data Visualization?

The process of transforming data

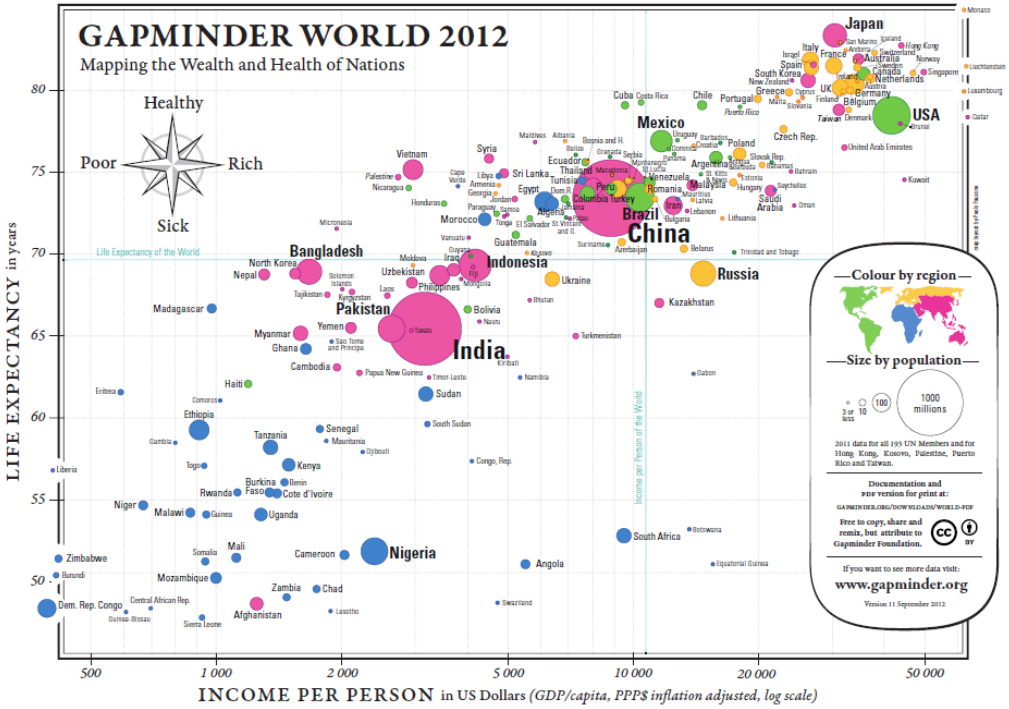
into visual representations

that reveal patterns, relationships, and insights

to support understanding, communication, and decision-making.

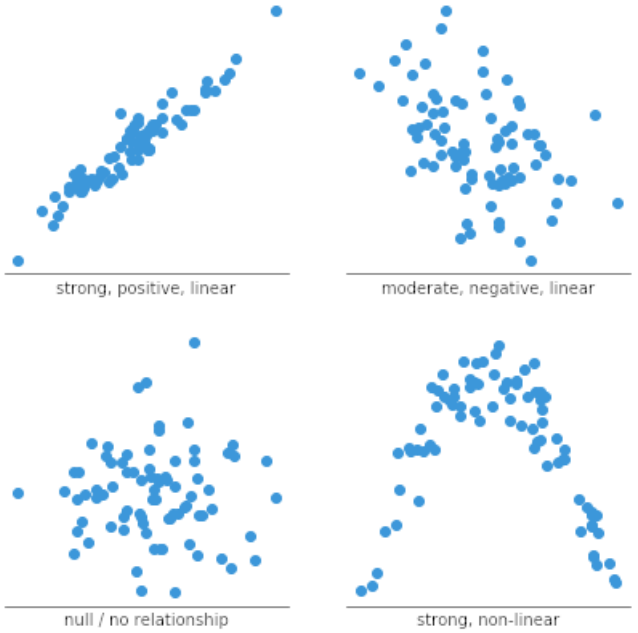
Raw data vs. visualization

country	region	income per person	life expectancy	population
Afghanistan	asia_west	1840	57.2	30700000
Albania	europe_east	10400	77	2920000
Algeria	africa_north	13200	76.8	37600000
Andorra	europe_west	41900	82.6	82400
Angola	africa_sub_saharan	6000	61.7	25100000
Antigua and Barbuda	america_north	19100	77	96800
Argentina	america_south	19200	76.1	42100000
Armenia	europe_east	7510	74.3	2880000
Australia	east_asia_pacific	42600	82.3	22800000
Austria	europe_west	44400	80.9	8520000
Azerbaijan	europe_east	15900	70.2	9260000
Bahamas	america_north	23000	73.7	372000
Bahrain	asia_west	41500	76.3	1300000
Bangladesh	asia_west	2710	71.3	156000000
Barbados	america_north	15400	76.8	282000
Belarus	europe_east	17500	71.8	9470000
Belgium	europe_west	41000	80.3	11100000
Belize	america_north	7970	71.6	337000
Benin	africa_sub_saharan	1860	62.6	9730000
Bhutan	asia_west	7030	72.9	753000
Thailand	east_asia_pacific	14400	77.2	67800000
Timor-Leste	east_asia_pacific	2030	72	1160000
Togo	africa_sub_saharan	1260	60.4	6860000
Tonga	east_asia_pacific	5130	70	105000
Trinidad and Tobago	america_north	31300	72.8	1340000
Tunisia	africa_north	10400	77.1	10900000
Turkey	europe_east	20300	78.6	74600000
Turkmenistan	asia_west	12200	68.8	5270000
Uganda	africa_sub_saharan	1640	58.6	36300000
Ukraine	europe_east	8320	71.1	45300000
United Arab Emirates	asia_west	59800	76.4	8900000
United Kingdom	europe_west	36700	80.7	64300000
United States	america_north	50500	78.9	313000000
Uruguay	america_south	18500	76.5	3400000
Uzbekistan	asia_west	4770	69.3	29500000
Vanuatu	east_asia_pacific	2900	63.3	247000
Venezuela	america_south	17700	75.3	29900000
Vietnam	east_asia_pacific	4910	73.6	90500000
Yemen	asia_west	3790	67.9	24900000
Zambia	africa_sub_saharan	3510	54.5	14700000
Zimbabwe	africa_sub_saharan	1850	54.1	14700000

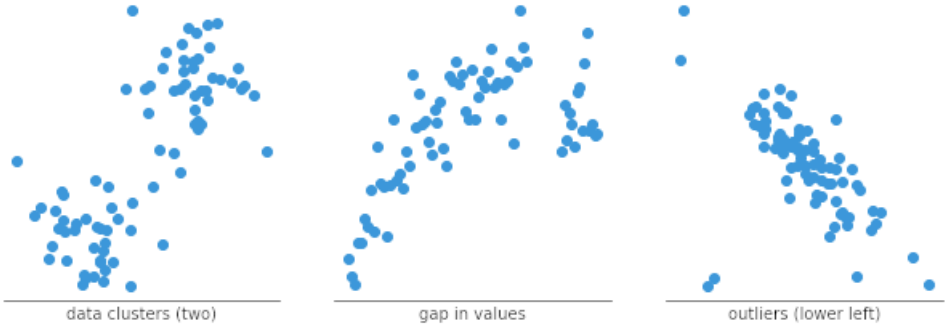


What a simple visualization can do

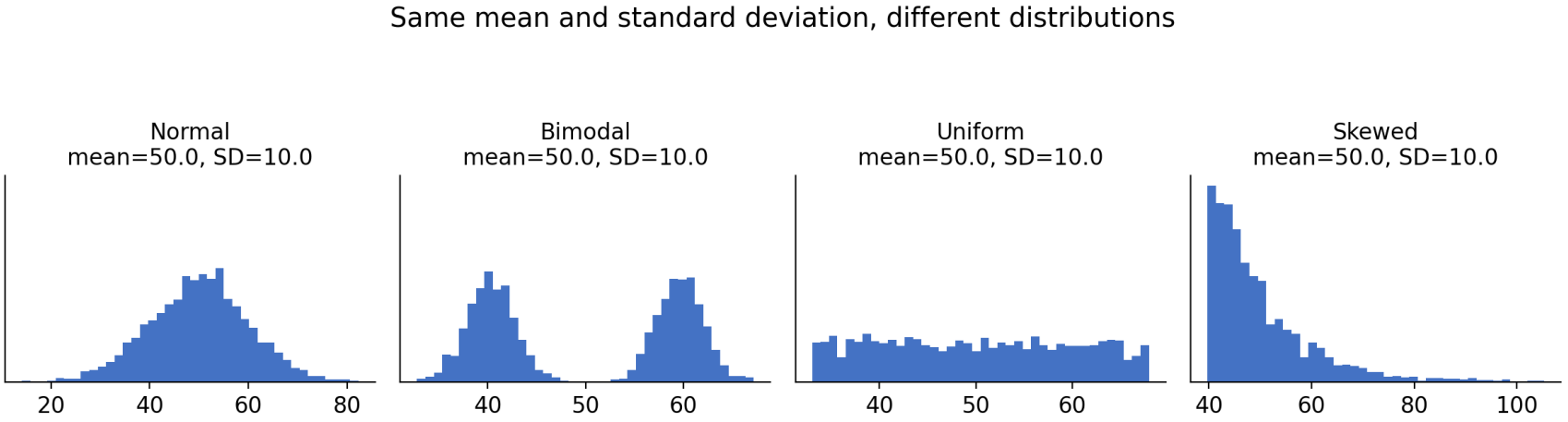
A scatter plot can reveal correlational relationships



and other patterns



Same statistics, different pictures

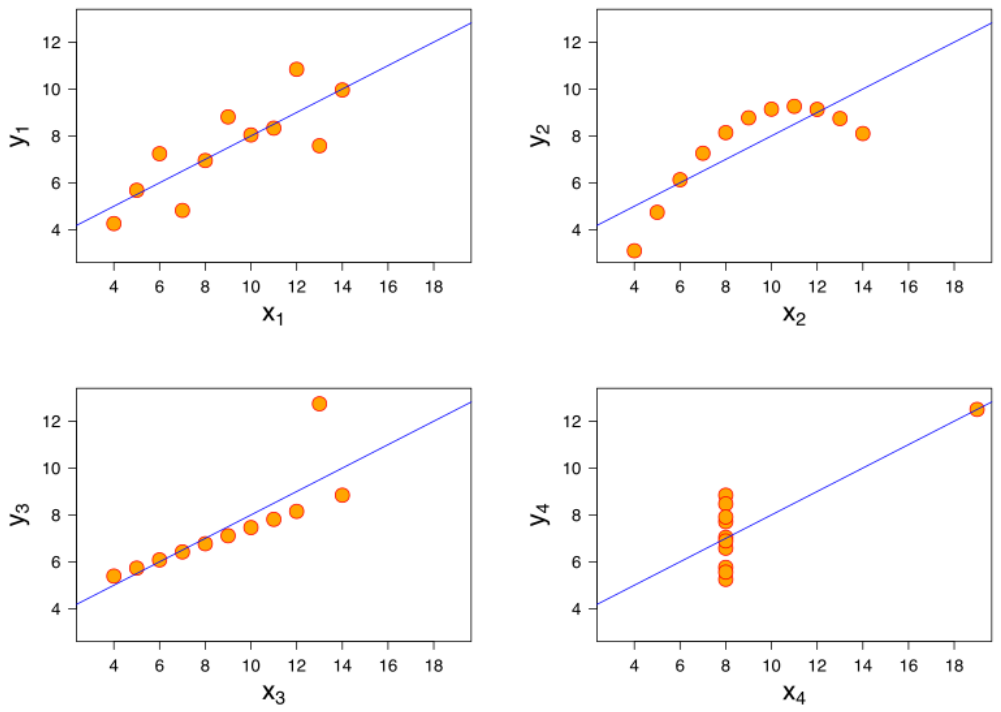


Anscombe's quartet

- Demonstrate the importance of visual representation of data
- All four datasets have almost identical statistics but different graphs

I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

Property	Value
Mean of *x*	9
Sample variance of *x*	11
Mean of *y*	7.50
Sample variance of *y*	4.125
Correlation between *x* and *y*	0.816
Linear regression line	$y = 3.00 + 0.500x$
Coefficient of determination of the linear regression	0.67

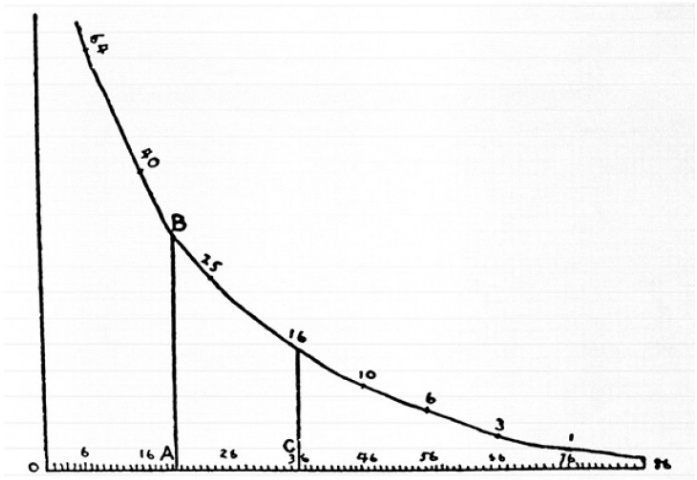


https://en.wikipedia.org/wiki/Anscombe%27s_quartet

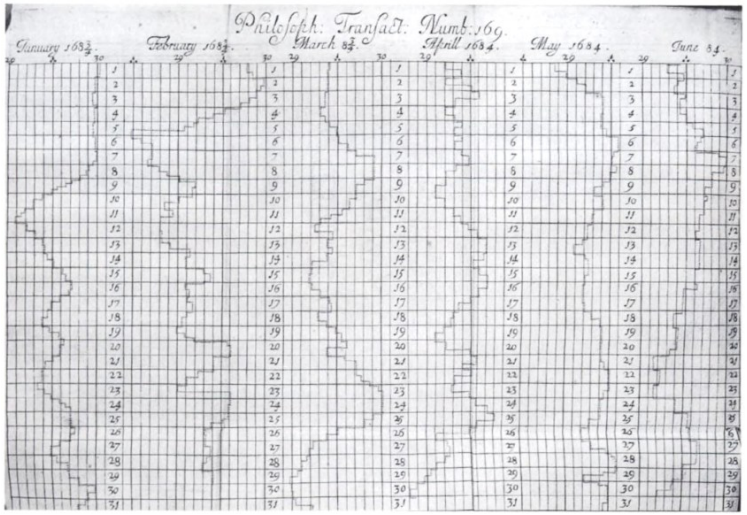
Four types of visualization

- **Relational graphics**: relationships between variables (this lecture)
- **Time series**: change over time (this lecture)
- **Spatial data maps**: data tied to locations (Lecture 2)
- **Network graphs**: relationships between entities (Lecture 3)
- These are broad families, not a strict taxonomy: they can overlap.

Where it began: four early visualizations



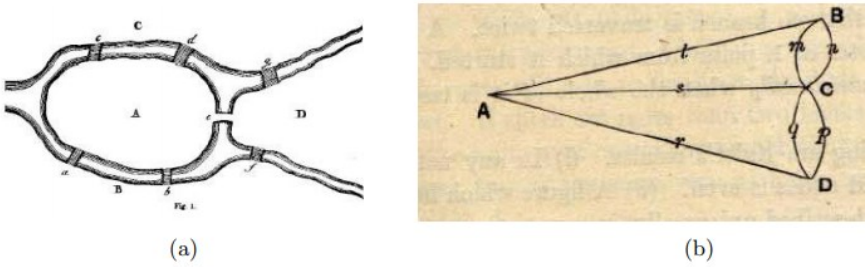
1669 Huygens: survival rate by age
(relational)



1685 Robert Plot: daily barometric pressure in Oxford
(time series)

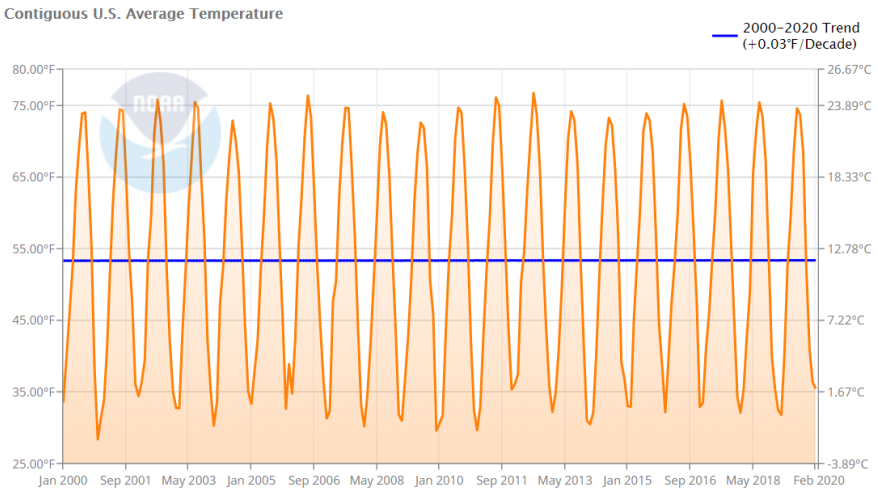


1686 Halley: trade winds and monsoons
(map)

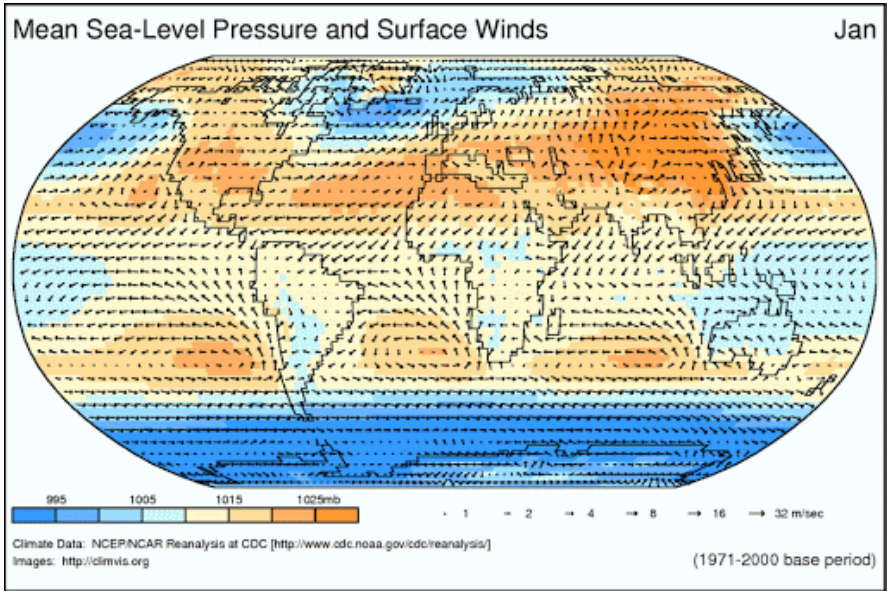


1736 Euler, 1892 Rouse Ball : the seven bridges of Königsberg
(network)

Time series and maps today



U.S. average temperature, NOAA [Climate at a Glance]



Mean sea-level pressure and surface winds (animated climatology)

Network Graph

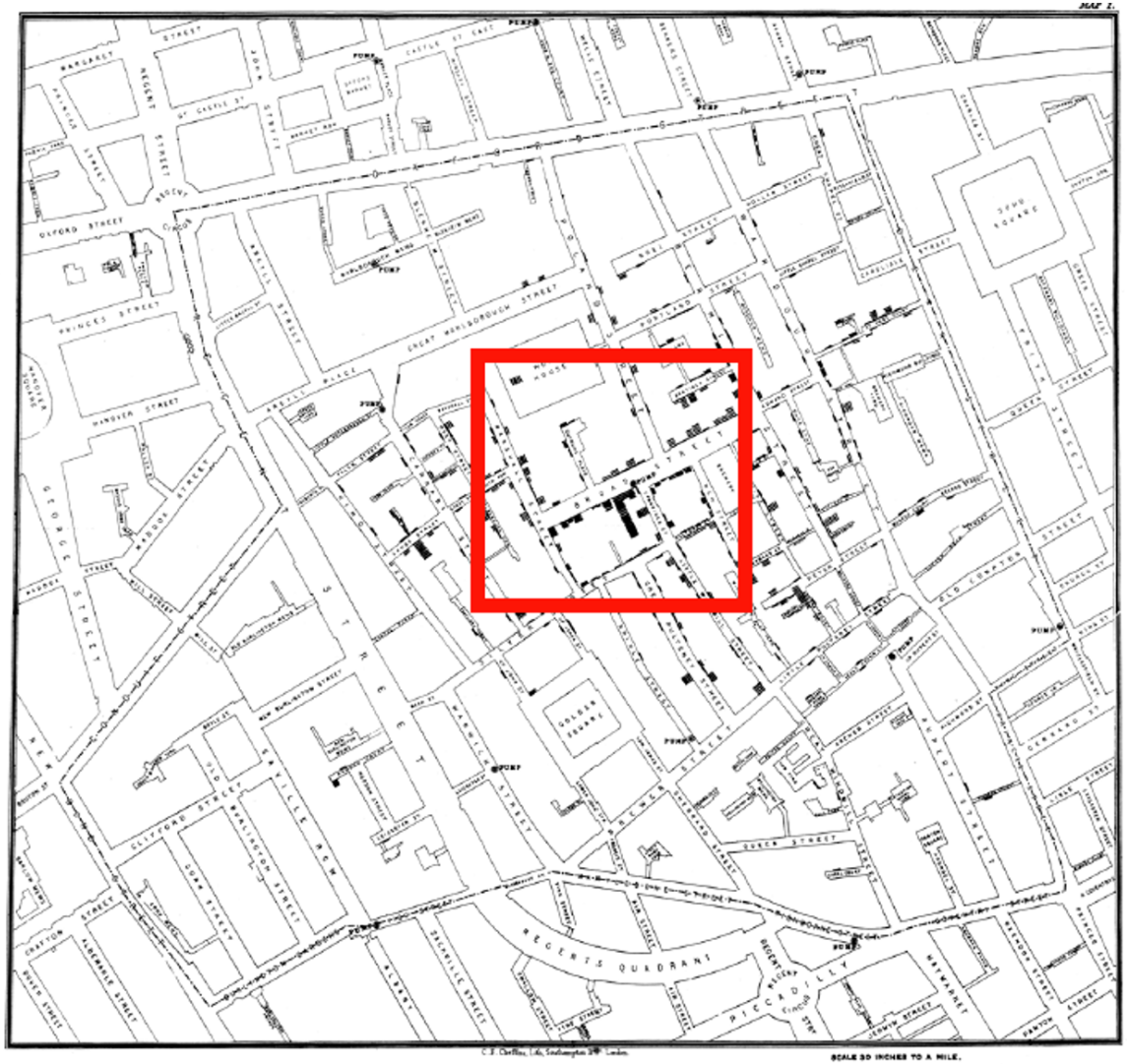
Facebook friendships (2010)

Each line connects between two cities, weighted by the number of friendships and geographic distance.



Paul Butler, Facebook Engineering, Dec 2010 https://www.facebook.com/notes/10158791468612200/

Classic Example: Dr.John Snow's Cholera Map (1855)



To stop the outbreak of cholera in London in 1854, **Dr. John Snow marked the cholera deaths on a map**. This map visualization indicated that the water from a pump on **Broad Street** was to blame as a large number of deaths were marked close to that pump. Snow's visualization is one of the most important early examples of epidemiology, that **clearly linked cholera's spread to water and not air**.

Snow, 1855 in *On the Mode of Communication of Cholera*

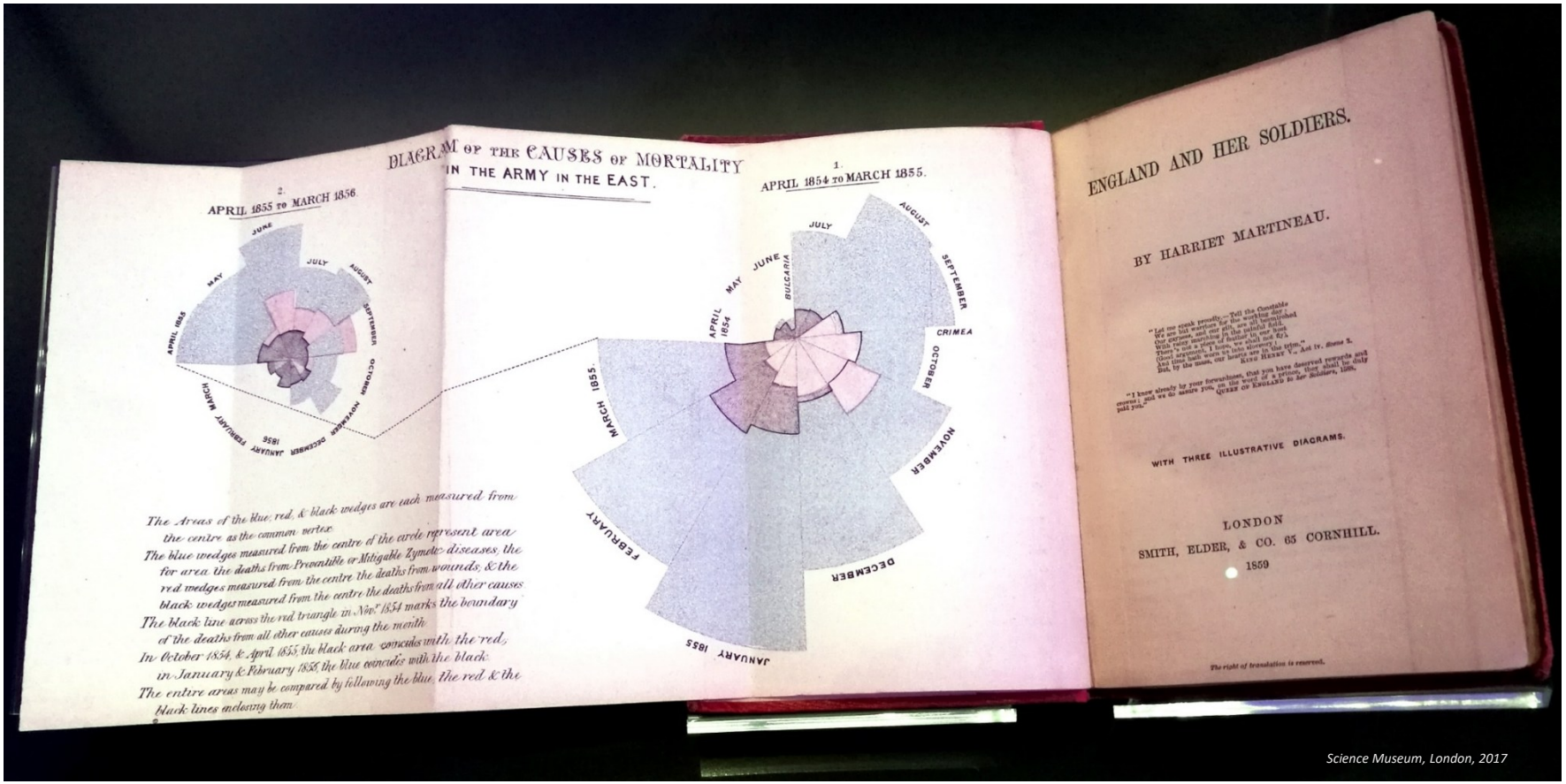
Classic Example: Dr.John Snow's Cholera Map (1855)



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Snow, 1855 in
On the Mode of Communication of Cholera

Classic Example: Florence Nightingale (1859)

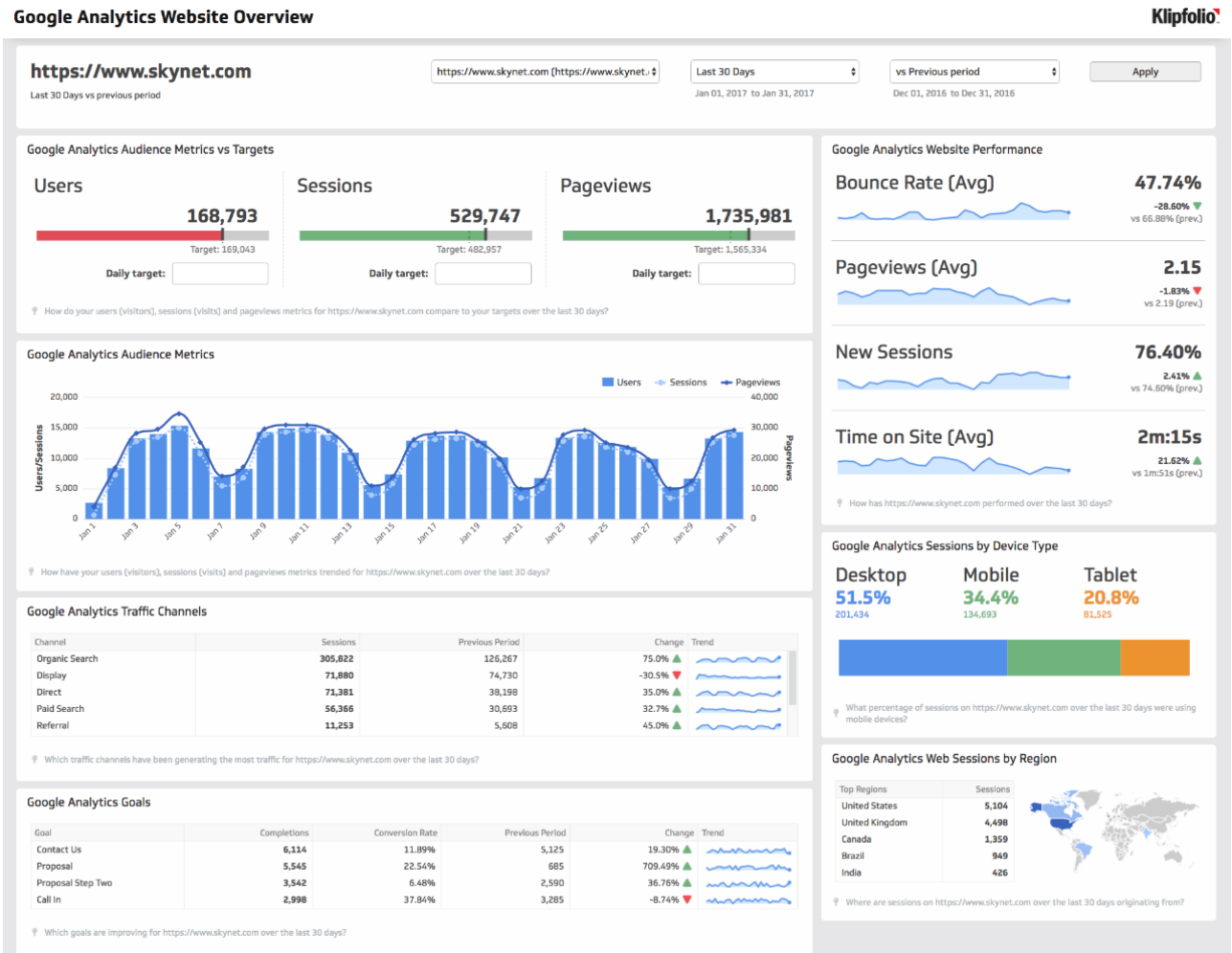


- Polar area diagrams (or coxcomb diagram) showed the causes of soldiers' deaths in each month.
- The right diagram showed the data of the first year when she arrived the hospital.
- Blue area showed deaths from preventable diseases due to bad conditions in hospital. Red area showed deaths from battlefield wounds. Black represented other causes.
- The left diagram showed the data of the second year after she implemented hygiene practices.

https://beta.sciencemuseum.org.uk/stories/2016/11/4/florence-nightingalethe-pioneer-statistician

Modern Example: Dashboard

Display many measures or key performance indicators (KPIs) on different charts.



Dashboards





Stock market technical chart

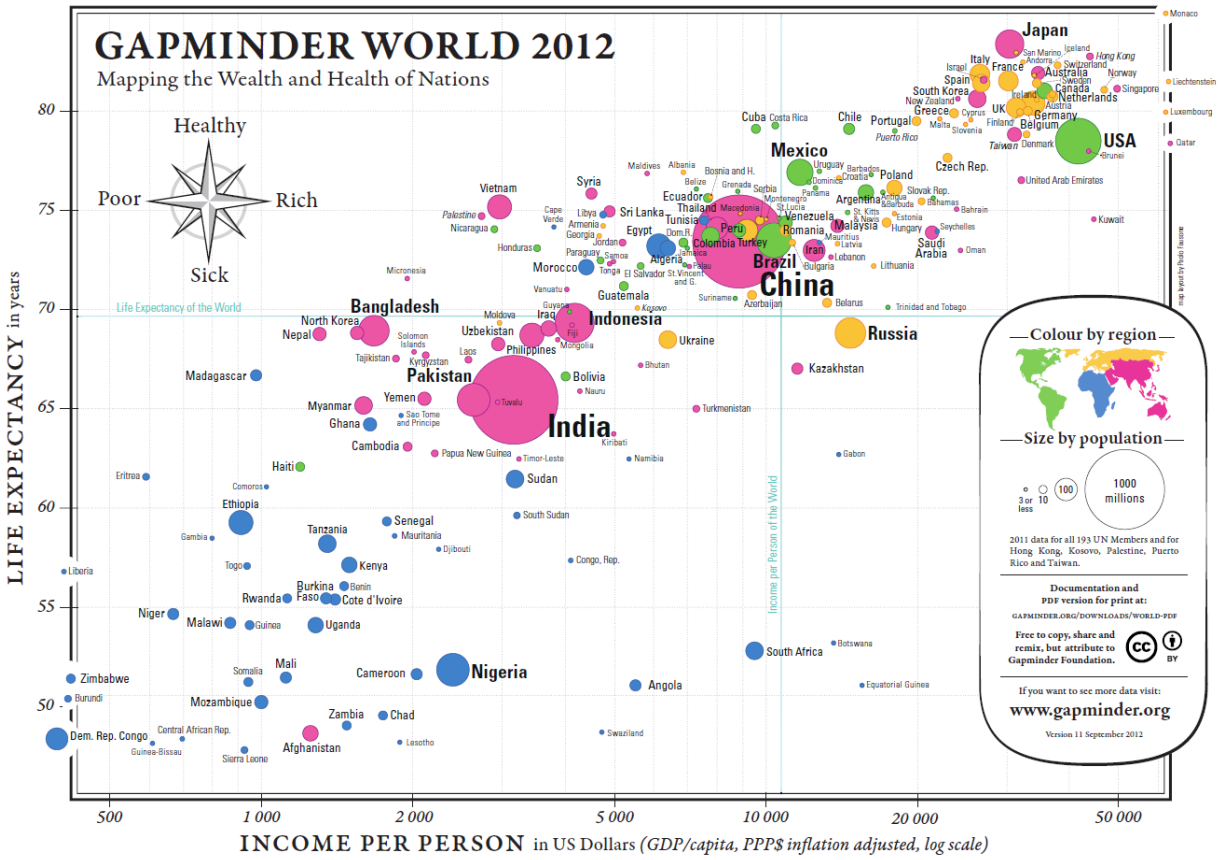


JHU COVID-19 dashboard

[Traffic congestion dashboard (Tableau)]

https://public.tableau.com/app/profile/veera.muangsin/viz/CongestionbasedonTravelTimeIndex/TravelTimeIndexTTIDashboard

Storytelling: Hans Rosling



"200 Countries, 200 Years, 4 Minutes" (BBC Four, 2010)

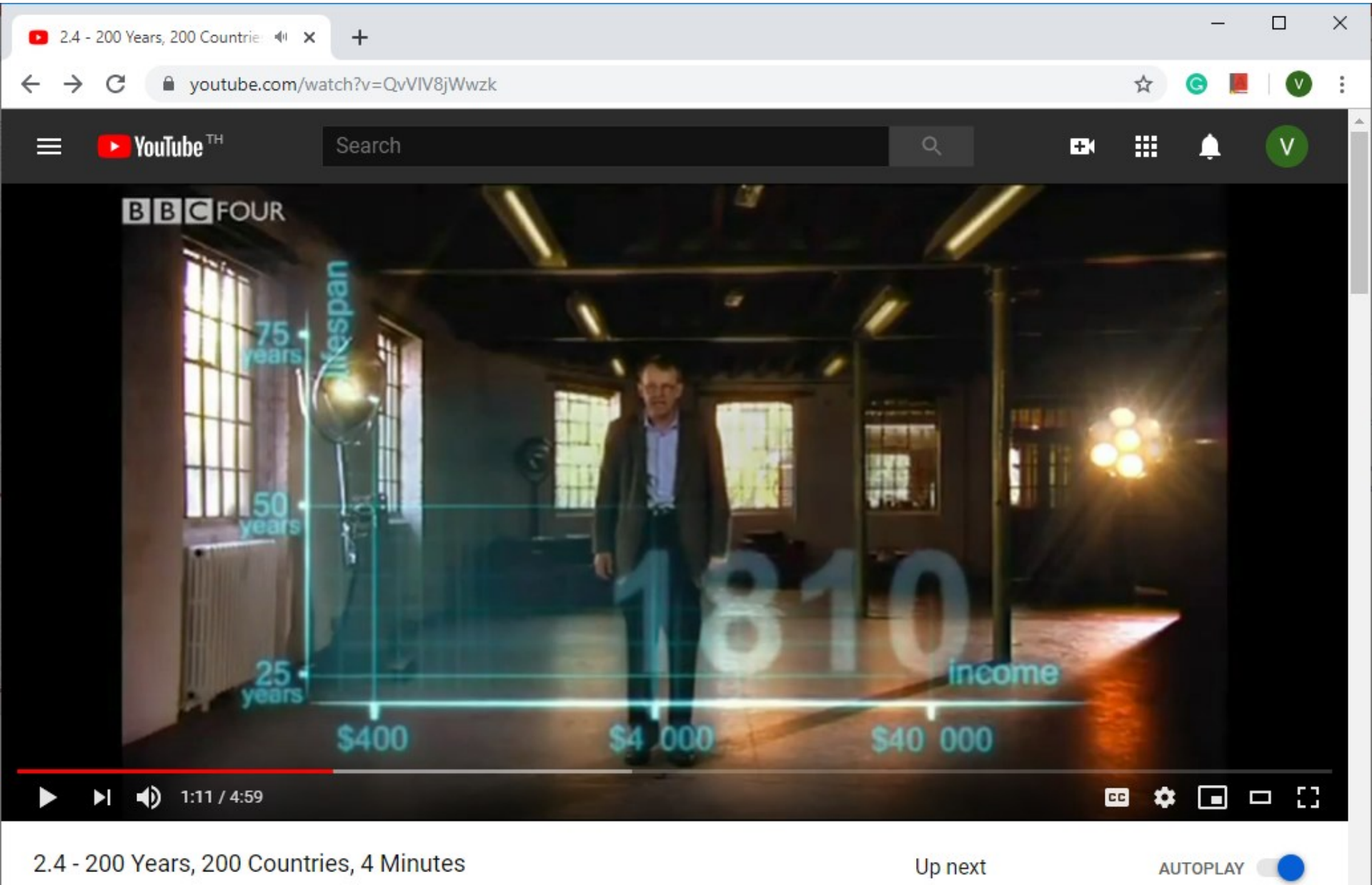
[Watch on YouTube]

Try it: [Gapminder Tools]

Five variables in one animated scatter: income (x), life expectancy (y), population (size), region (color), year (time).

Story Telling with Visualization by Hans Rosling

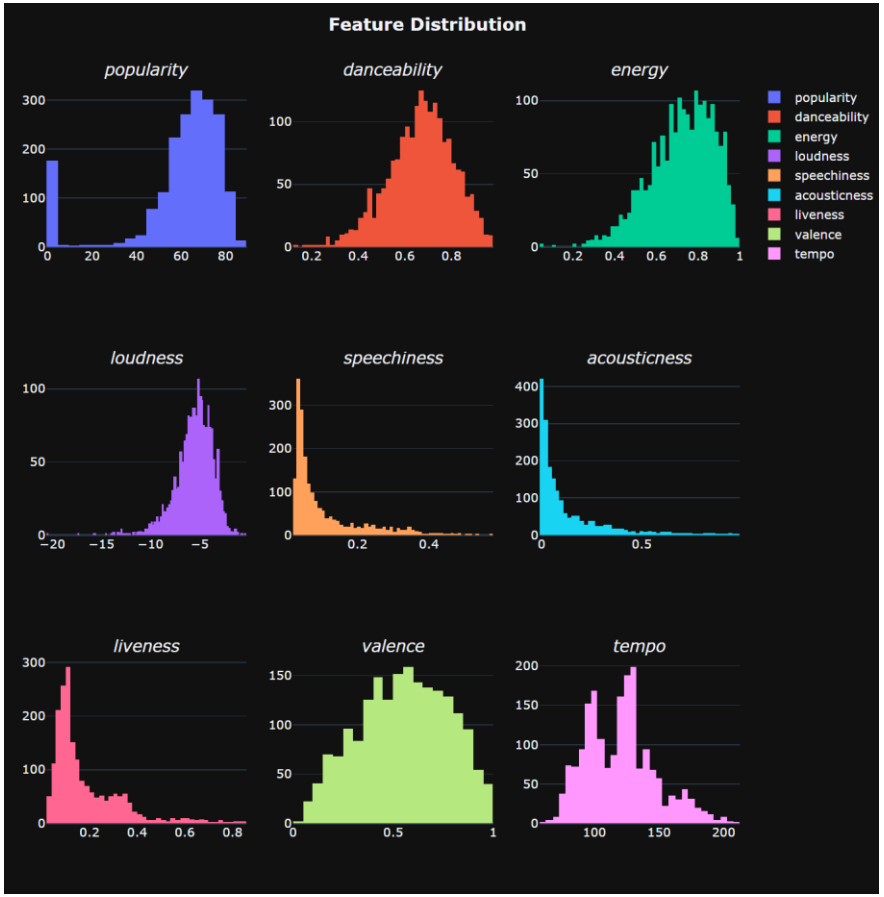
https://www.youtube.com/watch?v=QvVlV8jWwzk

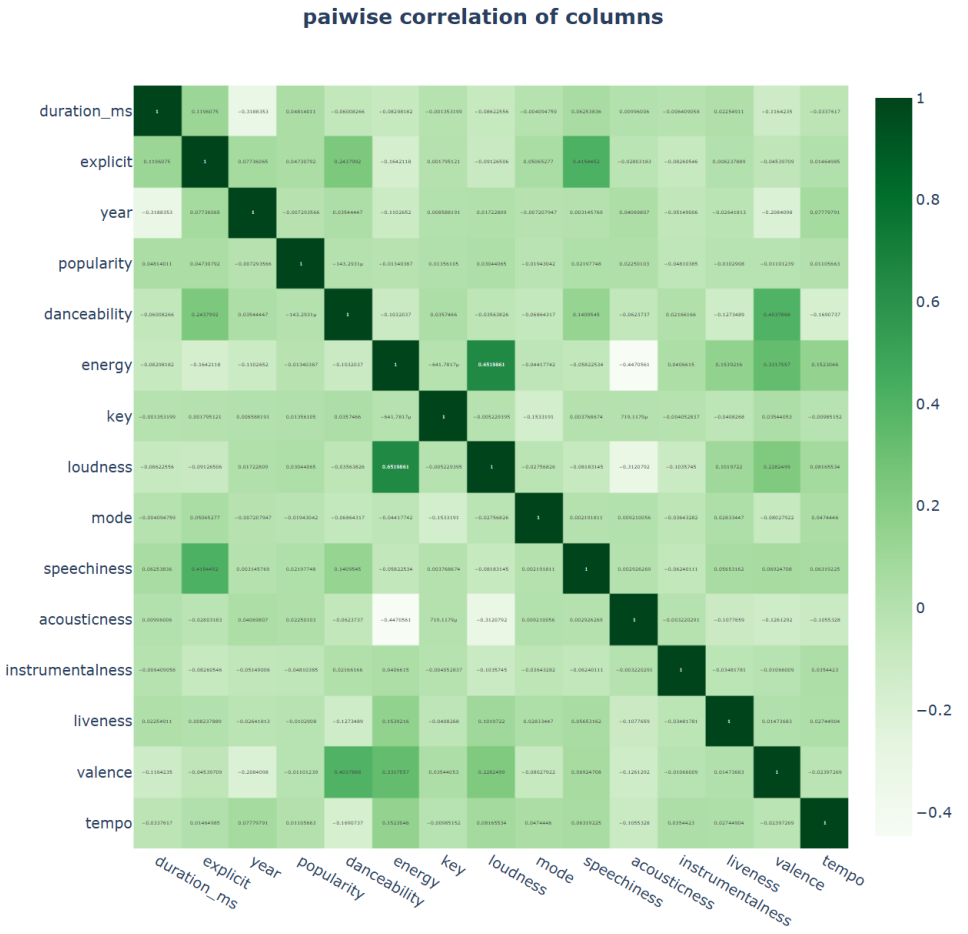


Example: Visualization in a data science project

Spotify Data Visualization

https://www.kaggle.com/code/varunsaikanuri/spotify-data-visualization





Principles of Effective Data Visualization

What is a good visualization?

- Not just a pretty chart. A good visualization helps the viewer:
  - understand the structure of the data
  - gain insight into the data
  - do tasks more effectively (present, explore, answer questions, decide)
- Is Dr. Snow's map good? Check it against all three.



Graphical excellence: less is more

Edward Tufte: Graphical excellence is well-designed presentation of interesting data,

- consists of complex ideas communicated with clarity, precision, and efficiency,
- gives the greatest number of ideas in the shortest time with the least ink in the smallest space,
- and tells the truth about the data.



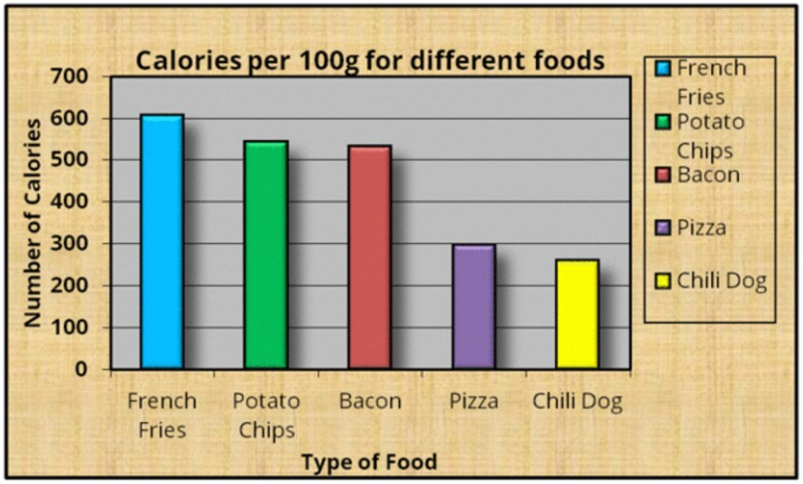
Remove to improve (the data-ink ratio). The author wants to highlight the data for bacon

Graphical excellence: less is more

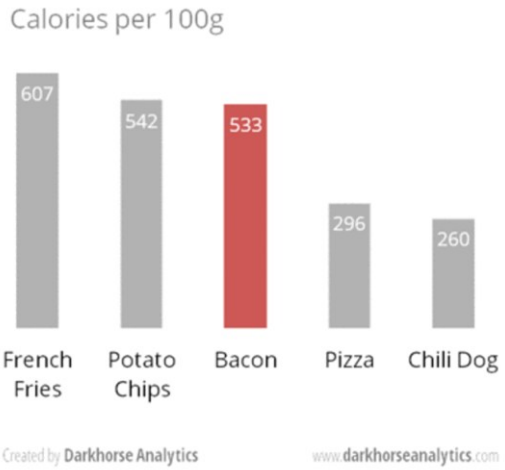
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Before

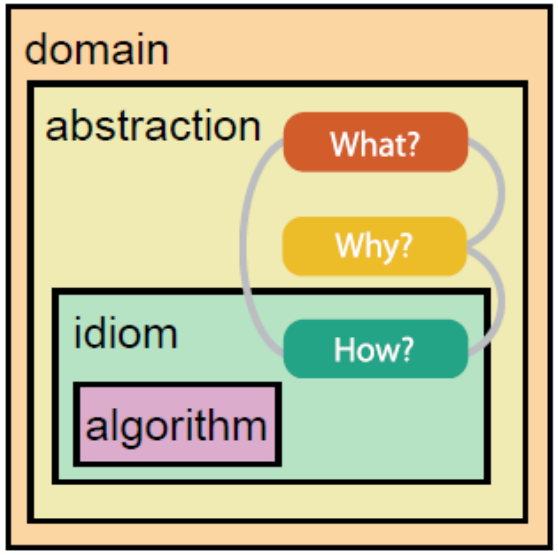


After



Remove to improve (the data-ink ratio). The author wants to highlight the data for bacon

Munzner's Visualization Design Methodology



- What is shown?
  - Data abstraction
- Why is it shown?
  - Task abstraction
- How is it shown?
  - Visual encoding

What: know your data

Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box		7/17/07
32	7/16/07	2-High	Medium Box		7/18/07
32	7/16/07	2-High	Medium Box		7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66		5-Low	Wrap Bag	0.56	1/20/05
69		4-Not Specified	Small Pack	0.44	6/6/05
69		4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
96	4/17/05	2-High	Small Box	0.55	4/19/05

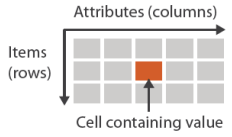
attribute

item

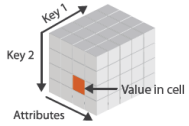
cell

Dataset Types

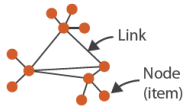
Tables



Multidimensional Table



Networks

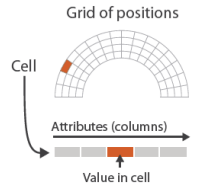


Trees



Spatial

Fields (Continuous)



Geometry (Spatial)



Attribute Types

Categorical

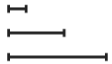


Ordered

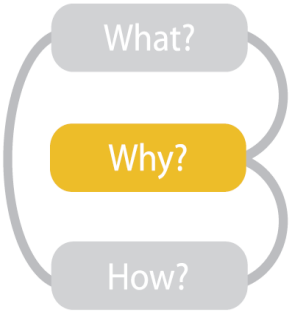
Ordinal

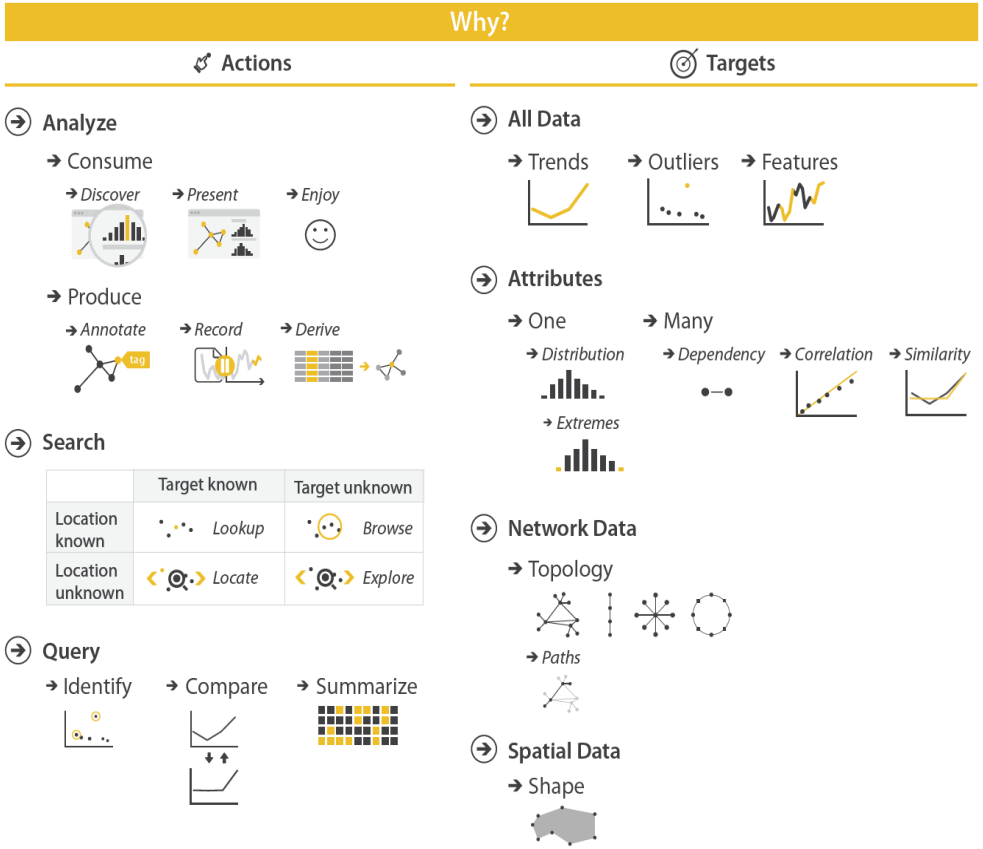


Quantitative



Why: tasks = {action, target}





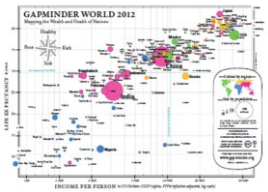
- {action, target} pairs
  - *discover distribution*
  - *compare trends*
  - *locate outliers*
  - *browse topology*

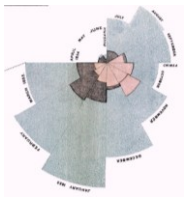
Examples

Gapminder: identify a correlation between income and live expectancy

Nightingale: compare distributions by cause of death

Snow: discover the spatial distribution of deaths

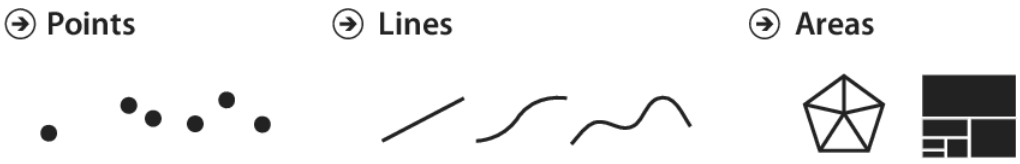




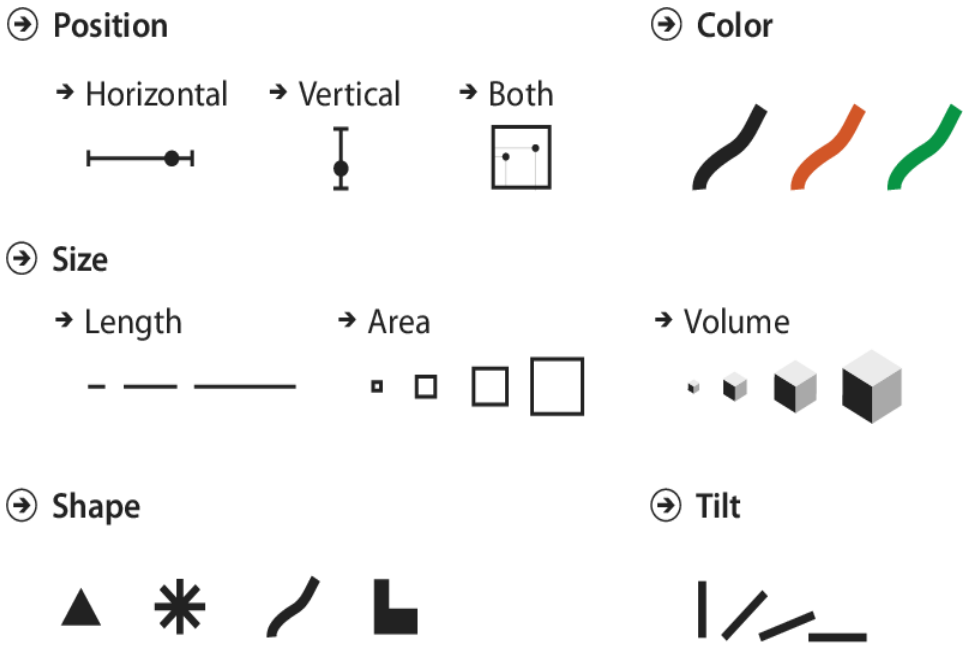


Visual Representation

Marks
Geometric primitives used to construct any visualization

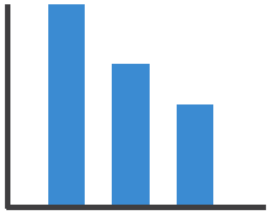


Visual Variables
The properties or appearance of marks that can be used to represent the data



Visual Encoding

- Combination of marks and visual variables



1:
vertical position

mark: line



2:
vertical position
horizontal position

mark: point



3:
vertical position
horizontal position
color hue

mark: point



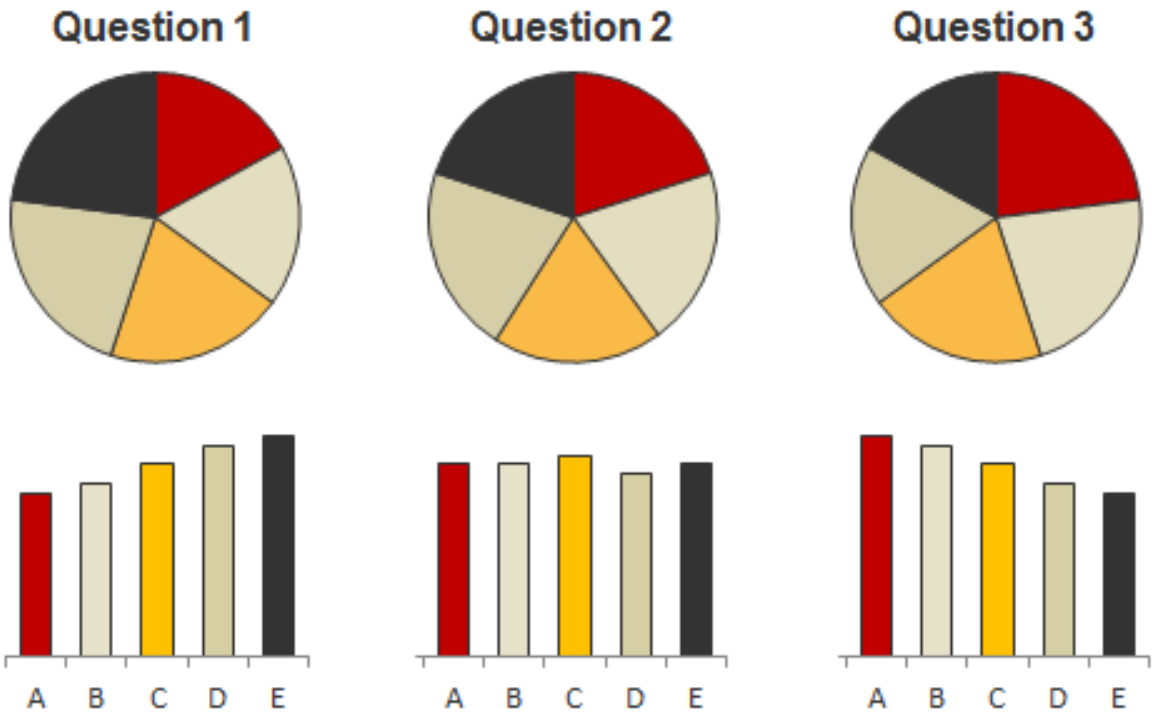
4:
vertical position
horizontal position
color hue
size (area)

mark: point

Effectiveness of Visual Variables

Pie Chart vs Bar Chart

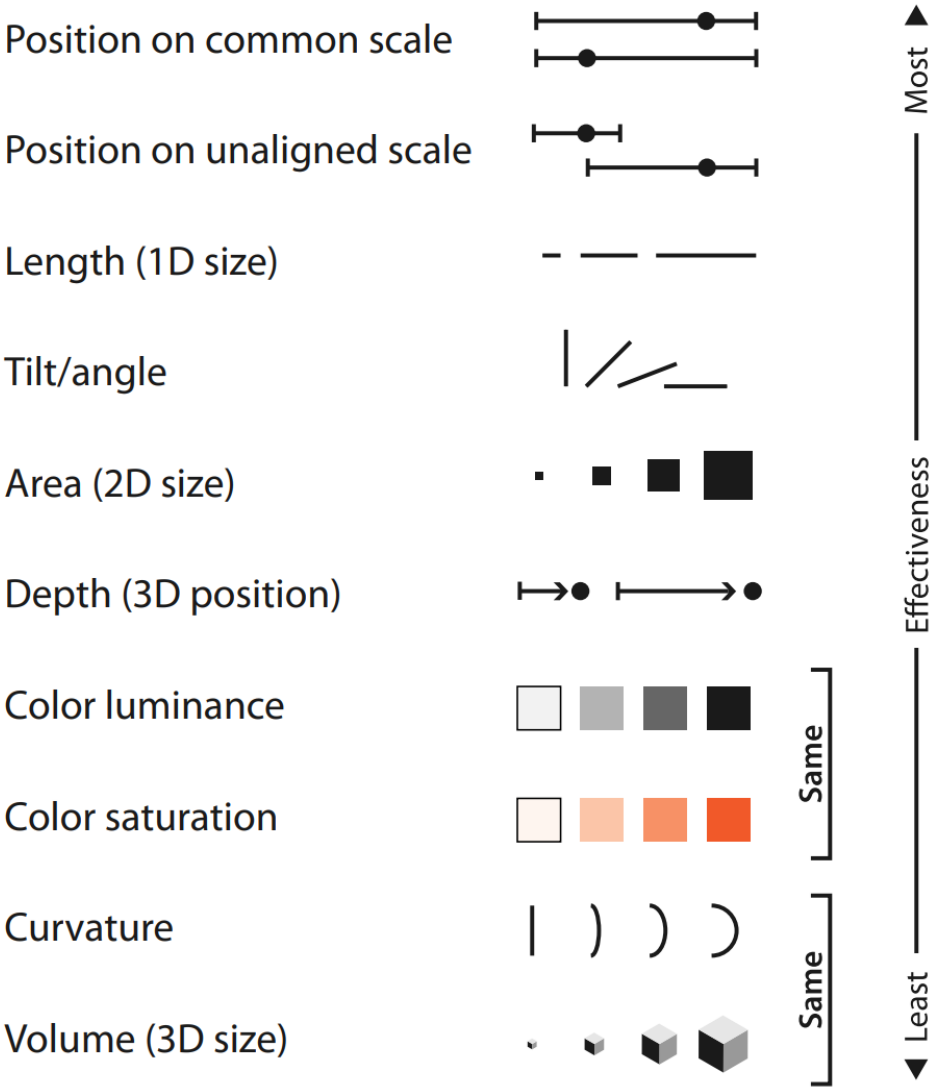
Which one is better? Why?



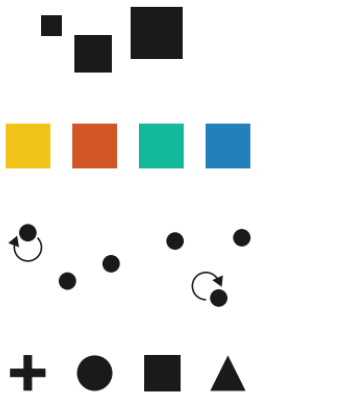
Effectiveness of a visual variable depends on type of data

Channels: Expressiveness Types and Effectiveness Ranks

Magnitude Channels: Ordered Attributes

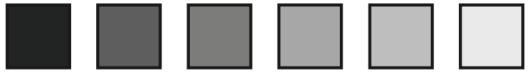


Identity Channels: Categorical Attributes

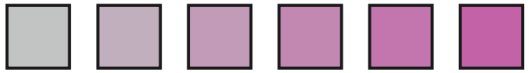


Color and order

Luminance



Saturation

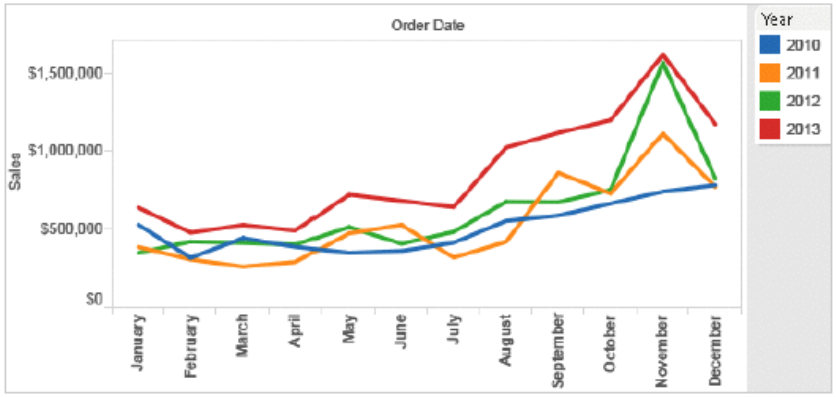


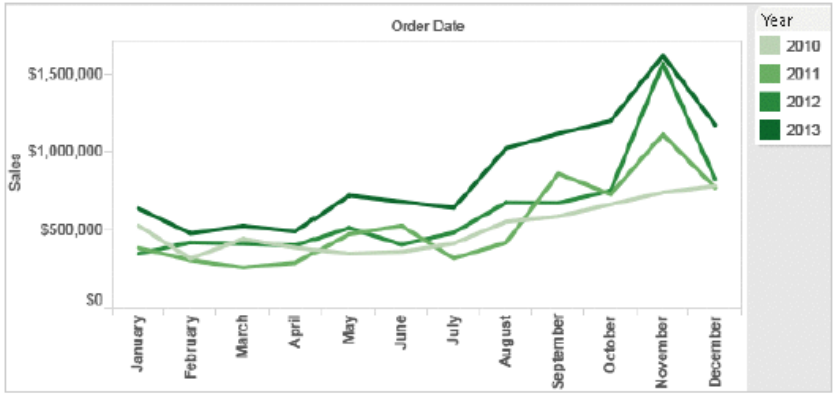
Hue



- **Hue** is unordered: use it for categories
- **Luminance / saturation** are ordered: use them for quantities

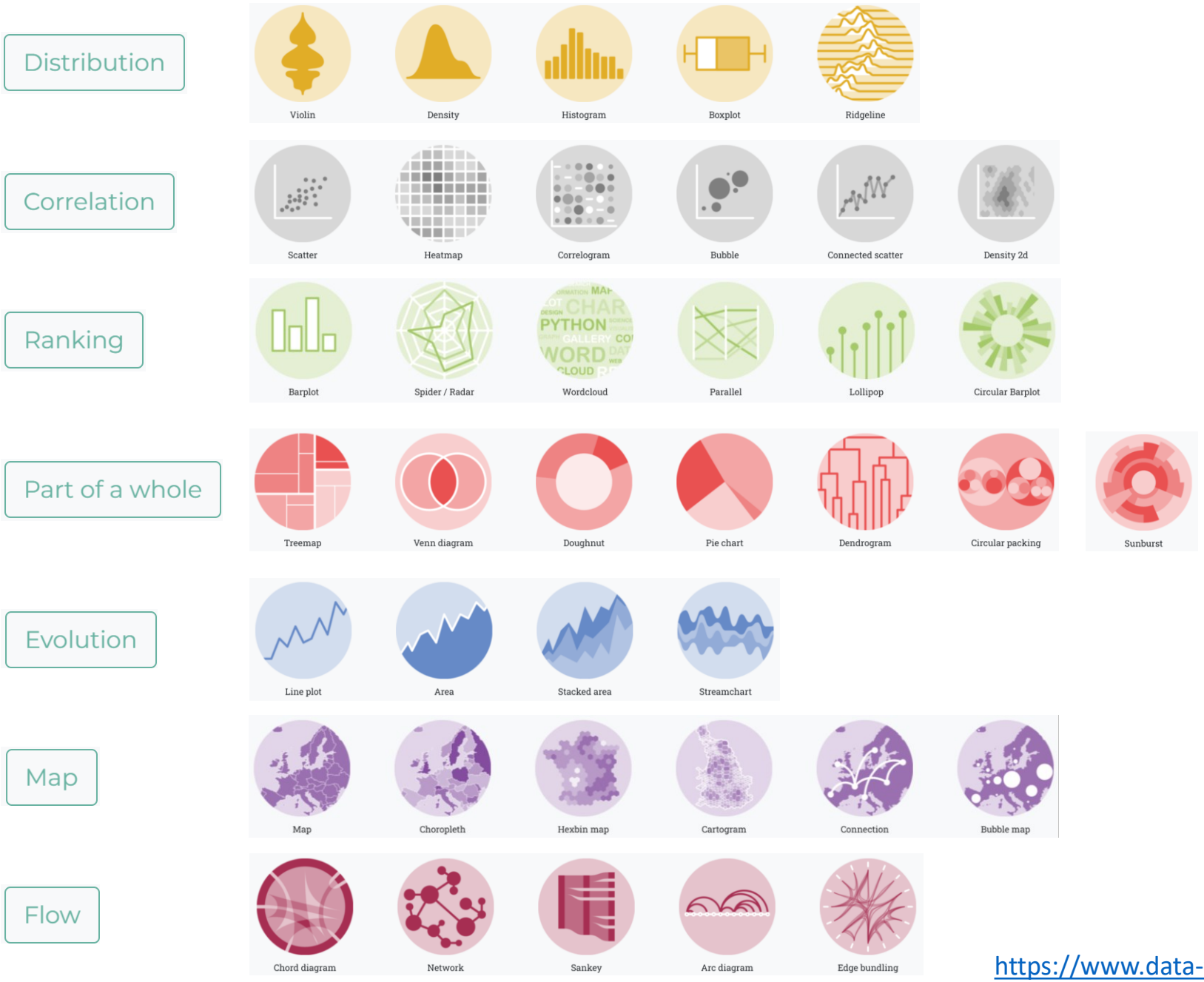
Categorical vs ordered color



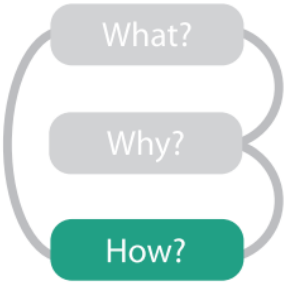


Choosing the Right Chart

Chart Types



https://www.data-to-viz.com/



How to select chart type

Based on

- Data [What is shown?]
- Task [Why look at it?]
- Effectiveness of visual variables
- Other factors (e.g. scalability)



Idiom: **Bar Chart**

marks: lines

visual variables:
length for quantitative value,
each mark separated horizontally, aligned vertically and ordered by label or length

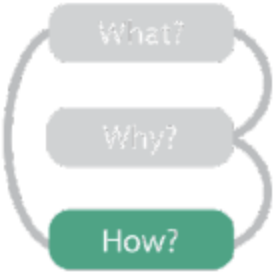
data: table with 1 category attribute (key attribute) and 1 quantitative attribute

tasks: compare, lookup values

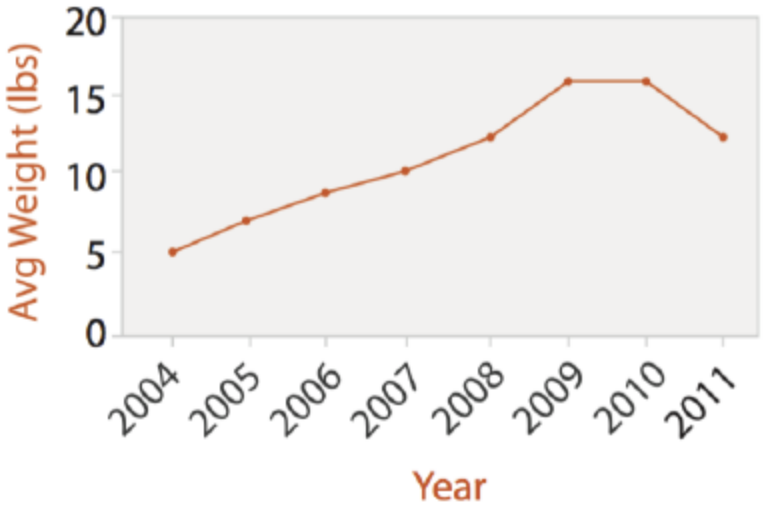
scalability: dozens to hundreds of levels for key attribute







Idiom: **Line Chart**

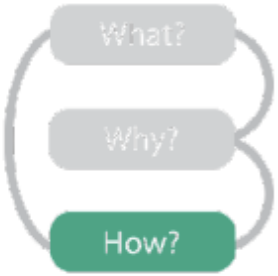


marks: points (a line connects the marks)

visual variables:
aligned length for quantitative value, horizontally separated and ordered by key attribute

data: table with 1 ordered attribute (key attribute) and 1 quantitative attribute

tasks: find trend (line connecting the marks emphasizes the ordering of the items along key axis)



Idiom: **Scatterplot**

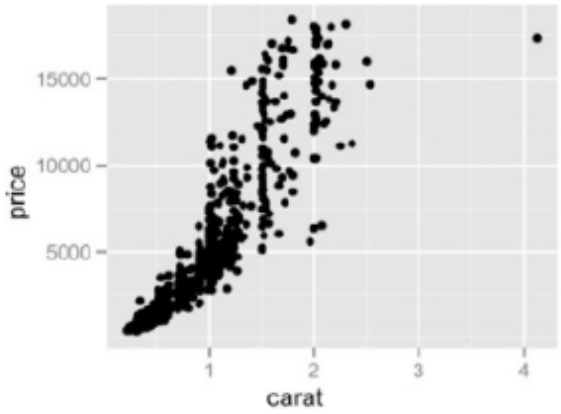
marks: points

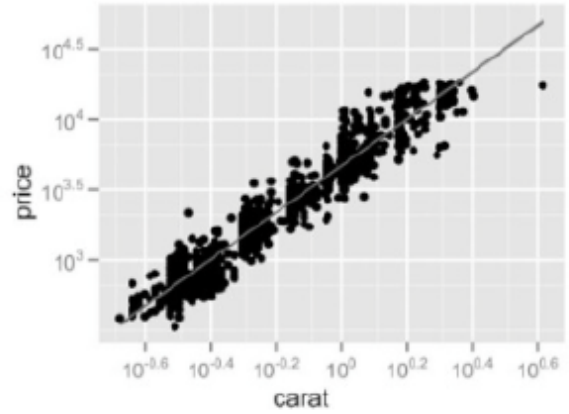
visual variables: position (horizontal + vertical)

data: table with only 2 quantitative attributes and no key (only values)

tasks: finding trends, outlier, distribution, correlation, clusters

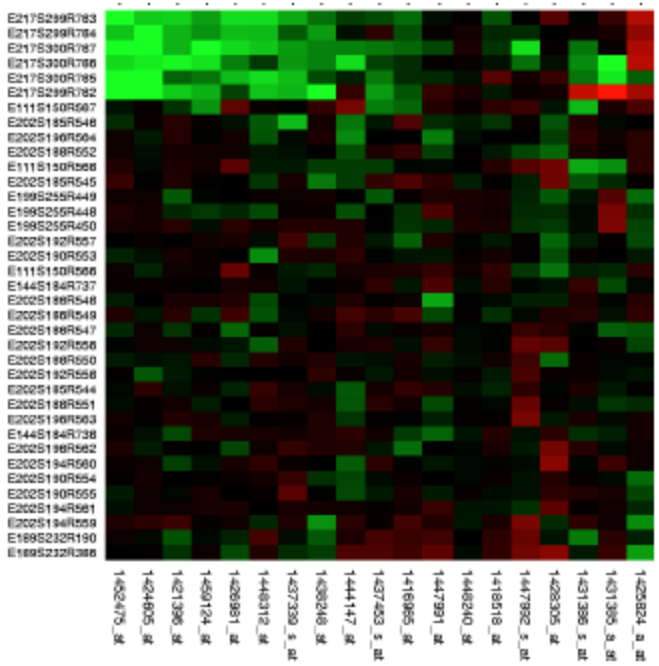
scalability: hundreds of items







Idiom: **Heatmap**



marks: area (separated and aligned in 2D matrix and indexed by 2 categorical attributes)

visual variables:

color for quantitative attribute

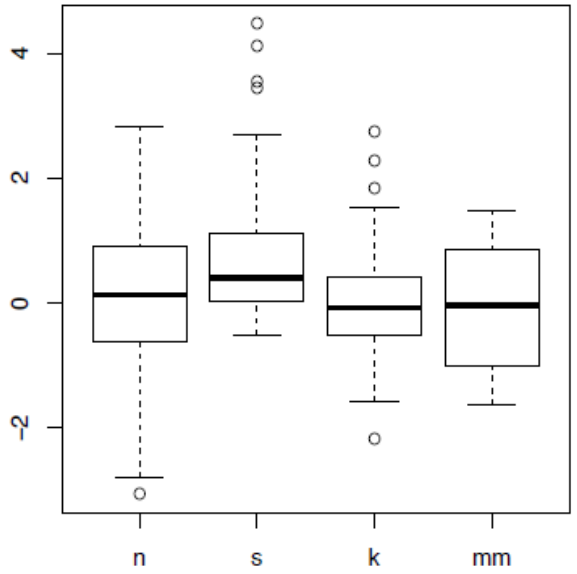
data: table with 2 categorical attributes (key attributes) and 1 quantitative attribute

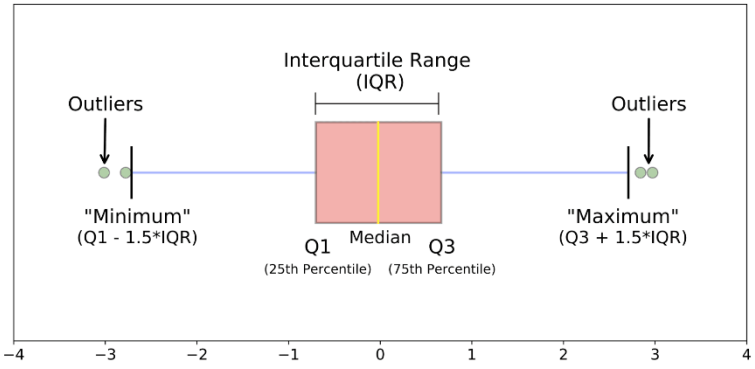
tasks: find clusters, outliers

scalability: 1 million items, hundreds of categorical levels, ~10 quantitative attribute levels

Idiom: **boxplot**

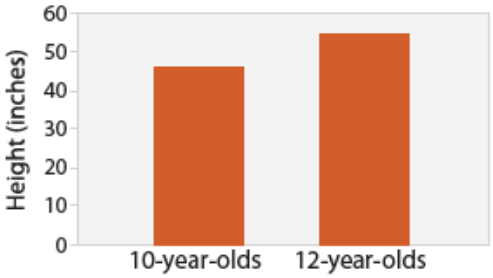
- static item aggregation
- task: find distribution
- data: table
- derived data
  - 5 quant attribs
    - median: central line
    - lower and upper quartile: boxes
    - lower upper fences: whiskers
      - values beyond which items are outliers
  - outliers beyond fence cutoffs explicitly shown

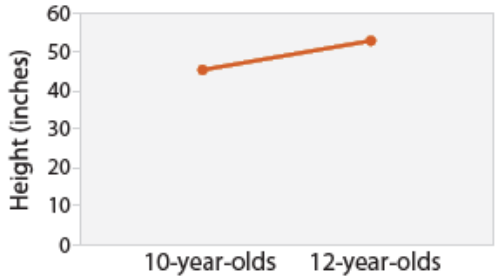


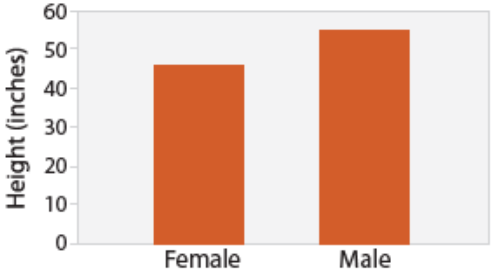


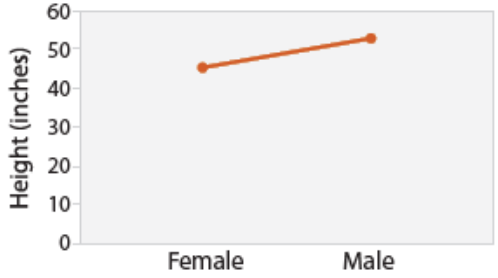
Choosing bar vs line charts

- depends on type of key attrib
  - –bar charts if categorical
  - –line charts if ordered
- do not use line charts for categorical key attribs
  - –violates expressiveness principle
    - implication of trend so strong that it overrides semantics!
      - – "The more male a person is, the taller he/she is"

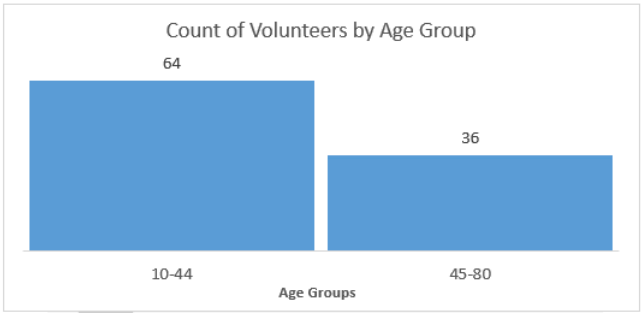


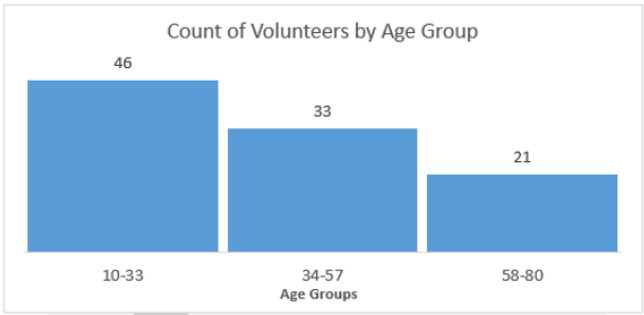


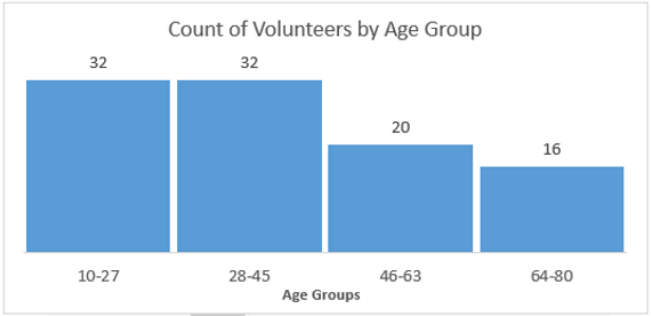


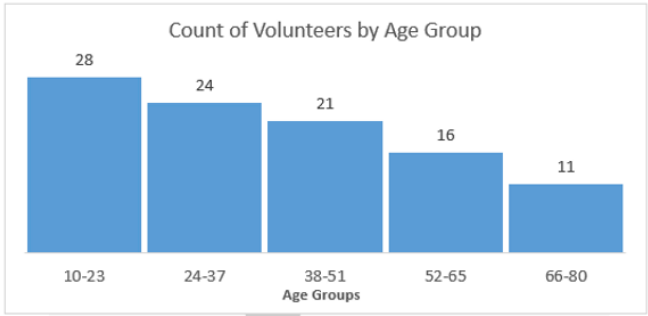


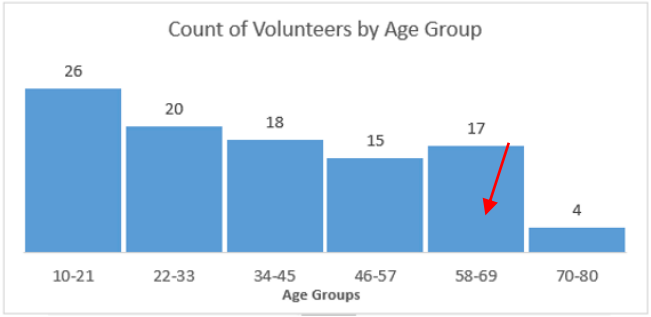
Histogram: different bin sizes tell different stories

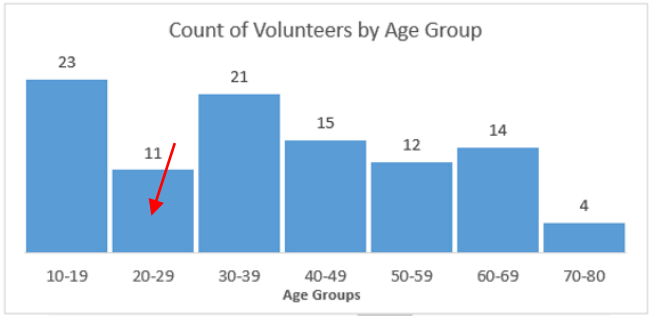






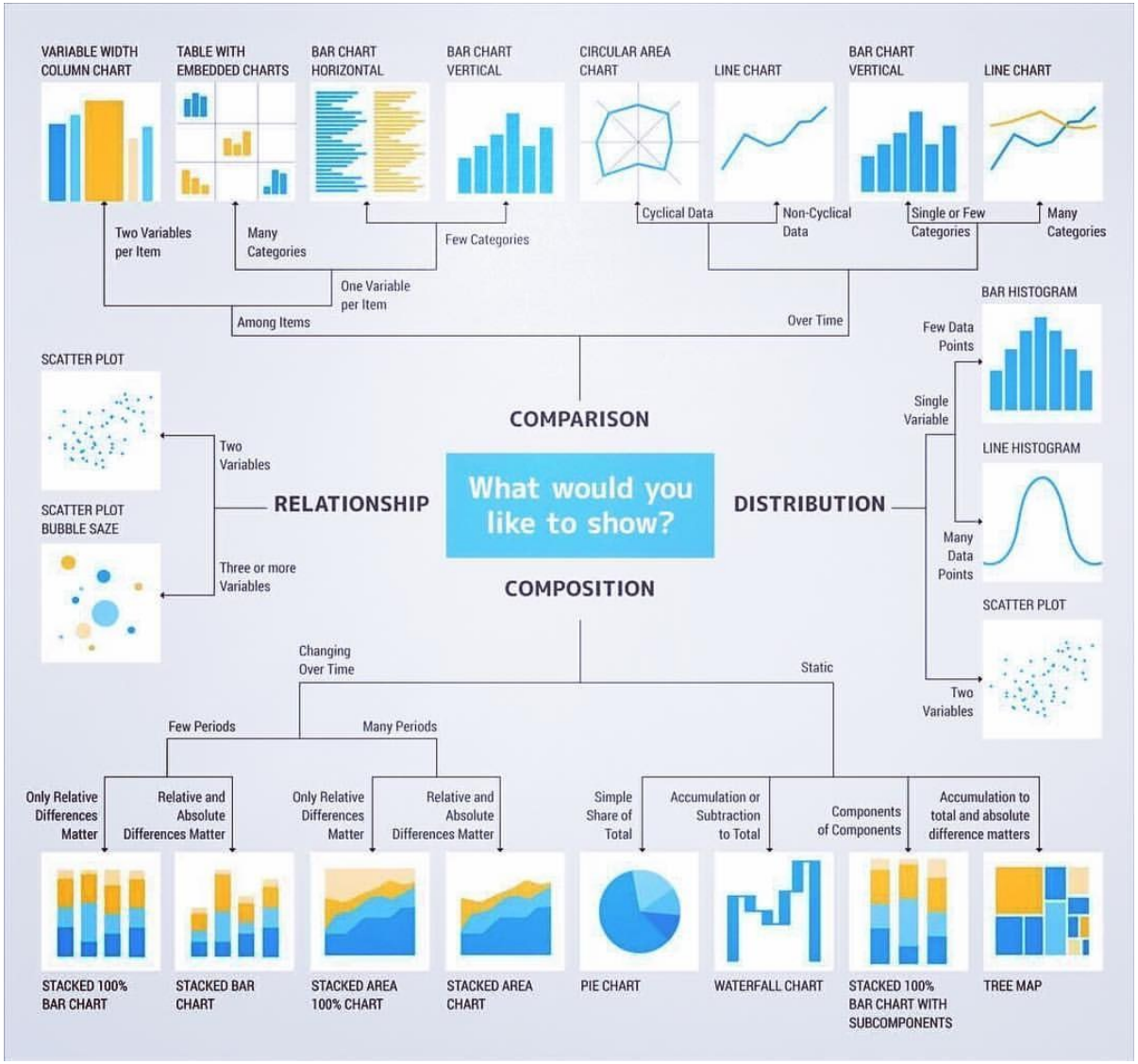






https://www.excelcampus.com/charts/dynamic-histogram/

Chart choosers



Start from the data type and the task, then pick a chart.

data-to-viz.com: decision tree from data type to chart, with Python code

Chart chooser (left): what would you like to show? comparison, relationship, distribution, composition

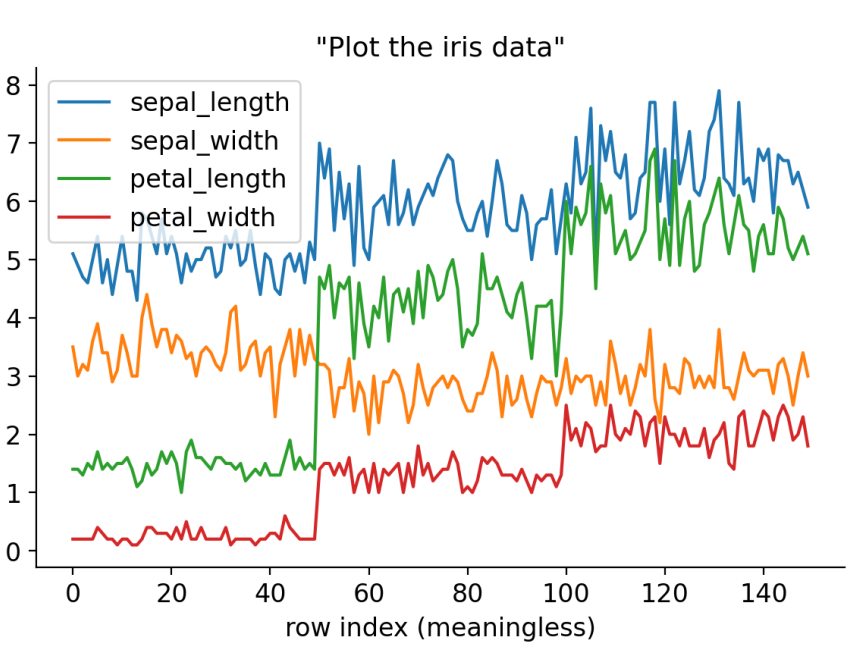
Data Visualization Tools

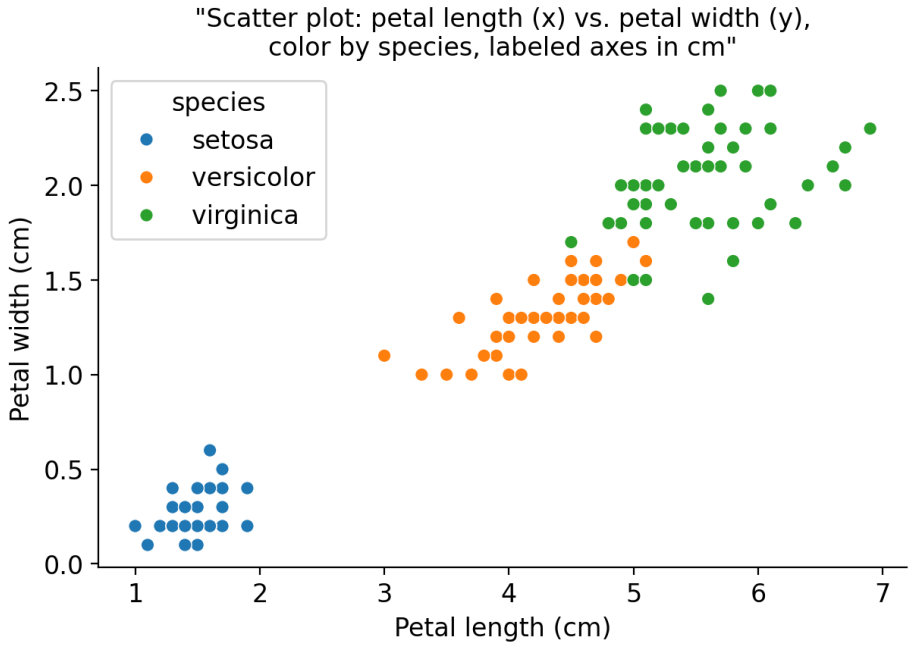
Type	Examples	Specialty
Dashboard	PowerBI, Tableau, Looker Studio (aka Google Data Studio)	
	Grafana	monitoring
Software package	ArcGIS, QGIS	map
	Gephi	network
Programming libraries	Python matplotlib, plotly, streamlit	
	Python streamlit, dash	web app
	Python folium, pydeck	map
	Python networkX, scikit-network, pyvis	network
	Javascript D3	
	R ggplot	

Visualization in the AI era

- AI assistants now write plotting code and draw a chart in seconds.
- What AI cannot decide for you:
  - Which question is worth asking
  - Whether the chart answers it (chart type, encoding, color, scale)
  - Whether the plotted numbers are right
  - What the chart hides, and what it does not prove
- The pattern for every example today: **question → chart choice → implementation → interpretation**
- By the end you can: choose and justify a chart, build it (yourself or with AI), check it, and interpret it cautiously.

What + Why + How = a good prompt





Vague prompt
“Plot the iris data.”

Result: a line chart of all columns against the row index. Correct code, meaningless chart.

What + Why + How
What: 150 flowers, 4 measurements (cm) + species.
Why: compare species by petal size.
How: scatter, petal length on x, petal width on y, color = species, axis labels with units.